PRE-HISTORY OF THE EARTH'S MAGNETIC FIELD

By E. A. Johnson, Thomas Murphy, and O. W. Torreson

Summary—In order to determine the origin and nature of the Earth's magnetic field and to test the various hypotheses which have been advanced to explain the field, it is desirable to determine the history of this field throughout geologic time and to investigate more carefully its spatial variations, both inside and outside the Earth's surface. This research is concerned with the determination of the history of the Earth's field as it can be deduced from the present polarization of crustal material. Unconsolidated fresh- and salt-water sediments have been investigated. These sediments are in the form of clays and offer one of the simplest types of polarization, since the clays can be redeposited under laboratory conditions. A particularly lengthy investigation of the polarization of glacial varves has been made, together with measurements on core samples of sediments from the Pacific. From a study of anomalous deposits in the glacial clays, the geologic stability of the polarization of these clays has been established over geologic time. From the measurements of the glacial clays, it is concluded that the Earth's field has not changed substantially in direction or intensity during the last 15,000 years. From measurements of the Pacific cores, it is tentatively concluded that the direction and intensity of the Earth's magnetic field has probably remained substantially constant during the last million years. A much more complete investigation is necessary to verify these tentative conclusions. It would be desirable to extend the measurements to periods of the order of one billion years. These results are consistent with the "fundamental" theory proposed by Schuster, Babcock, and Blackett, but do not provide positive evidence to support this theory.

Statement of the problem

The origin of the Earth's magnetic field remains a fundamental mystery. It is difficult to obtain adequate experimental data to describe the field sufficiently well so that a test can be made of the various hypotheses regarding the nature and origin of this field. There are two general types of theories which have been advanced to explain the magnetic field of the Earth. One set of theories assumes that the origin of the field is fundamentally connected with the mass and rotation of the Earth and may be an inherent property of all rotating matter. The second general class of theories attempts to explain the origin of the Earth's field on the basis of complex phenomena within the Earth. In order to test the validity of these hypotheses it is desirable to have a more complete knowledge of the variation of the field of the Earth with time and also to describe more fully the variation of the field from deep within the Earth to several Earth diameters. The arguments leading to the conclusion that the field of the Earth and other large bodies may be connected with angular momentum has been re-summarized recently by Blackett [see 1 of "References" at end of paper] in view of the new astronomical evidence obtained by Babcock [2] of Mount Wilson Observatory. The theories of the second type have been summarized

by Vestine, Tuve, and Johnson [3]. The work of Elsasser [4] is one of the most recent and promising of the second class of theories. Recent experimental evidence relating to the various theories has been discussed in England and summarized by Chapman [3*a*].

The purpose of the research reported in this paper is to provide as many data as possible on the time-variation of the Earth's magnetic field in both intensity and direction in order to provide experimental criteria with which to test the many hypotheses that have been advanced to explain the origin of the Earth's magnetic field. A reversal in direction of the field or a very large variation in intensity with time (confirming the change of four per cent per century recorded in the past few hundred years) would be regarded as evidence disproving the theory of Schuster and Blackett. The massive size of the Earth and its high internal conductivity make it impossible for the magnetic field of the Earth to change appreciably within historic time. To be useful, the period for which the history of the field must be determined is of the order of geologic time. It would be highly desirable to have this history for the last billion years.

Much work has been done by previous investigators on the study of the magnetization of baked clays and of rocks; the literature contains hundreds of names and goes back more than 250 years. Since 1890, particularly, investigations have been numerous, and Koenigsberger, in 1938, besides reporting on his own studies of the magnetization of eruptive rocks, has furnished an extensive bibliography [5] of the works of others. Among the more prolific of the early workers mentioned in this bibliography are G. Folgheraiter and P. Mercanton, whose investigations were made chiefly between 1890 and 1920 and were concerned with the remanent magnetization of old pottery and similar objects of baked clay, as well as with that of eruptive rocks.

B. Brunhes and P. David in 1901 [6] reported the results of some of their investigations on the magnetization of clay beds which had been baked to a kind of natural brick by overstreaming lava. This work, as well as their studies of eruptive rocks, inspired many other investigators in the next few decades.

In 1925, R. Chevallier [7] published a paper on the magnetization of the lavas of Mount Etna and discussed the direction of the magnetic field in Sicily from the Thirteenth to the Twentieth Century. The values of declination ranged from 20° east to 20° west, and indicated a period of secular variation in declination of approximately 750 years. The last 350 years of the declination-curve obtained from the work on lavas agreed excellently with known surface measurements with ordinary magnetometers; his additional data extended the variation back about 400 years more.

Results on the measurements of glacial varves from New England and

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deep-sea cores from the Atlantic were reported by Johnson and McNish [7*a*] in 1938. These provided a series of measurements of declination dating back 12,000 years, but evidence to confirm the geologic stability of the deposits was lacking.

The work of Chevallier interested Gustav Ising in Sweden sufficiently to induce him to begin a study of the magnetization of varved clays in that country. Work in 1926, 1930-31, 1933, and 1940 was summarized in a paper in 1941 [8]. Continued improvement in instruments and observational procedures after the preliminary work of 1926, culminated in measurements in 1940 of 350 years of varves from Viby. For the oldest of these varves, declination was found to differ only 10° from the present value and for the youngest was found not to differ appreciably. Inclination for the oldest varves was 50° and for the youngest 30°. The result for declination was interpreted by Ising as showing that the remanent magnetization had not remained constant since deposition, but had followed, in the main, the direction of the Earth's field, with the agreement best in the youngest varves.

In 1938, E. Thellier [9] published an extensive report on his many years of study of the magnetization of baked clays in the form of old bricks and pottery, and of basaltic rocks. He concludes that the basalts do not give good results for the study of the direction of magnetization in past geologic time, but that the results on baked bricks and pottery going back to the year 1400 are satisfactory. Data for inclination from the latter have extended knowledge back beyond the period covered by direct measurements in a satisfactory way for the region near Paris.

One of the most recent and most comprehensive discussions of the magnetization of clays and rocks appeared in 1942 as the work of Prof. Dr. Hans Haalck [10]. He summarizes his conclusions in 14 points, and, besides outlining the causes of magnetism in crustal materials and how and under what conditions the magnetism varies, he remarks that drawing general conclusions about magnetic characteristics from particular rock samples must be done with the greatest caution, since the samples may be affected by local natural deviations. He further states that drawing conclusions about the direction of the Earth's field in former times on the basis of observed directions in rock samples is a very uncertain procedure. He considers also that only in very rare cases can measurements on burnt clay vessels and bricks or on certain sedimentary rocks be used for this purpose. He does not believe the supposition that the magnetic field was reversed in some past epoch is at all well supported. Several of his concluding paragraphs suggest his belief that differences of temperature in various parts of the Earth's crust play an important part in the magnetization of the crustal materials, and he ventures to remark that one might suspect that the cause of the secular variation of the geomagnetic elements should

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be sought in changes of temperature inside the Earth's crust at depths where the temperature is near the Curie point.

A preliminary study of the blue clay of the Boston basin (approximately 15,000 years old) in the United States was made by Benedikt in 1943 [11]. Using a vertical torsion balance, he could determine the direction of maximum susceptibility in the plane of bedding of the clay. The torque was determined as the function of the field in Helmholz coils in which the sample was centrally located, the field being varied up to values of plus or minus 1600 oersteds. This paper was primarily a description of the method of measurement, and the data were not discussed.

Measurements of the total intensity of the Earth's field from burnt clay bricks several hundred years old obtained near Lille and Paris, France, were reported in 1942 and 1946 by E. Thellier and O. Thellier [12, 13]. Values of 0.56 gauss at Lille and 0.66 and 0.71 gauss at Paris are considerably greater than the value of 0.45 gauss which now exists, pointing to the possibility that the intensity has decreased notably in the course of some hundreds of years. These results agree rather well with the decrease of about four per cent per century, shown by records obtained during the last few hundred years.

The present investigations, using new apparatus, have been concerned chiefly with the case of unconsolidated clays. For the most part, these clays have had a relatively simple thermal, chemical, and mechanical history. They can be redeposited in the laboratory and the mechanism of polarization can be studied. Thus they offer a favorable material from which the magnetic field at the time of deposition can be deduced.

Description of apparatus

The method employed to measure the polarization of the clay samples is the same as that described by E. A. Johnson [13*a*]. The apparatus is similar to his and will be described in detail in a forthcoming paper. Briefly, the sample is rotated close to the center of a coil and acting as a small magnetic dipole induces an alternating voltage in the coil. The intensity and phase of the amplified signal are measured, from which the magnetization of the sample can be deduced.

The sample in the form of a cylinder four cm in diameter and four cm long is rotated at the end of a shaft at ten cycles per second, being driven through a reducing gear by a standard synchronous motor. The coil, counter-wound to eliminate external alternating magnetic fields, is connected to the amplifier through a suitable transformer. The amplifier consists of eight stages, each resistance-capacity tuned to about ten cycles and strongly degenerated to ensure stability in operation. The overall gain of each stage may be chosen as ten or one in order that the final amplified signal shall have a value between five and ten volts. The tuning of the amplifier is so

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designed that the discrimination against 60 cycles as compared with ten cycles is 10^8. No interference has been encountered from external 60-cycle power circuits either in the laboratory or in the field.

The intensity of the amplified signal is measured with a vacuum-tube-

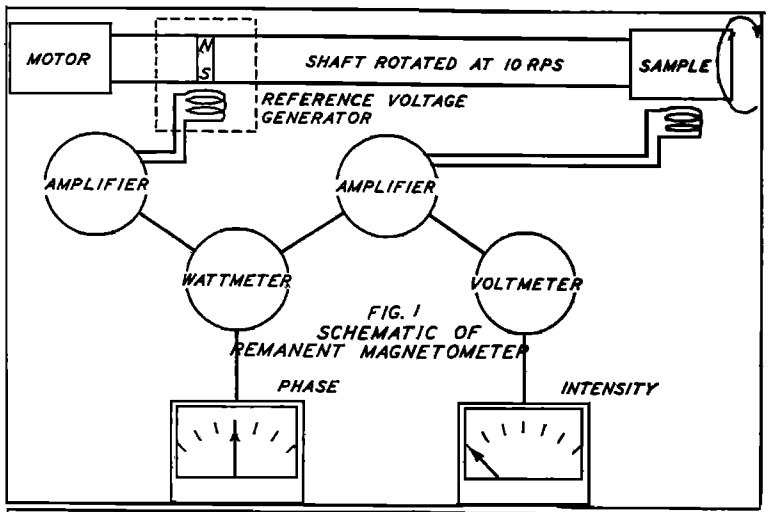


FIG. 1
SCHEMATIC OF REMANENT MAGNETOMETER

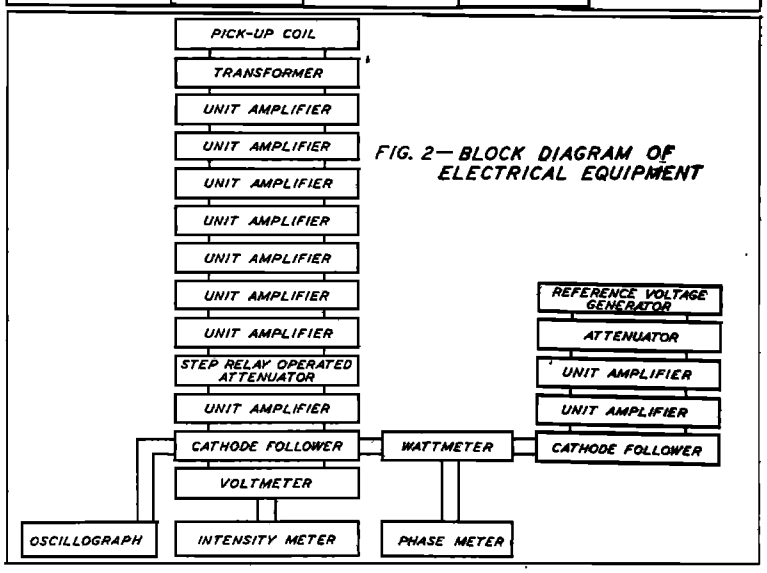


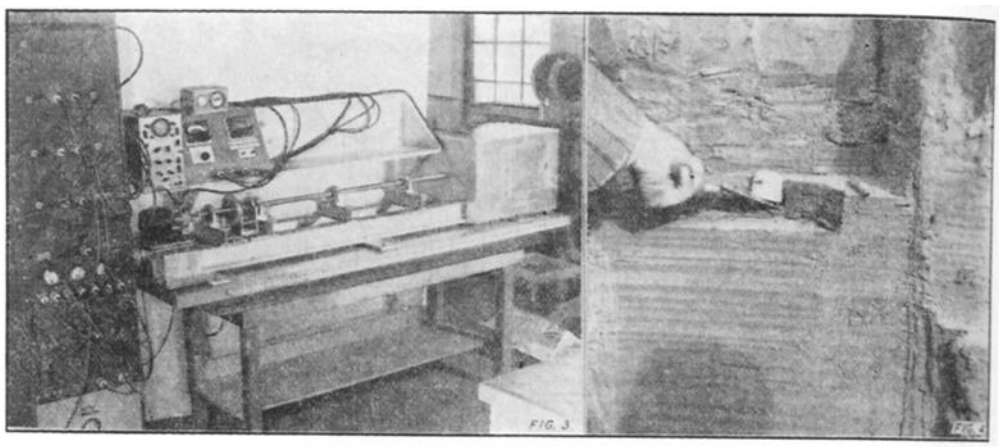
FIG. 2—BLOCK DIAGRAM OF ELECTRICAL EQUIPMENT

voltmeter circuit and its phase in a wattmeter circuit. The latter consists of a ring modulator circuit utilizing germanium diodes and is used as a null indicator. To balance the amplified signal from the sample, a signal is supplied from a magnetic generator mounted in the shaft of the apparatus.

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The phase of this signal can be changed through 360° and during measurement is adjusted to a value 90° out of phase with the signal from the sample to give zero reading on the wattmeter. The apparatus is calibrated in intensity and phase by using a small standard magnet in place of the sample. Figures 1 and 2 are schematic and block diagrams of the apparatus.

The apparatus, except for the electronic equipment, was constructed of non-magnetic materials and very ruggedly built to withstand vibration and shock during transport in a large laboratory truck which is taken to sites at which the clays and rocks are found. View of apparatus is shown in Figure 3.



The sensitivity permits a moment of 2×10^{-8} cgs unit per cc to be detected. For moments of 10^{-6} cgs unit per cc, the direction of magnetization can be determined to within one degree.

Methods of sampling

The clay samples discussed in the present paper were obtained from the glacial clays of New England at sites fully described by Antevs [14] following his work in 1922-26. He identified about 5000 layers or varves in the Connecticut Valley region and gives to the earliest layer, which he located near Hartford, Connecticut, the arbitrary date 3000. The latest varves were located in Vermont and dated about 8000.

Figure 5 shows the sites investigated during 1947. Many of Antevs' original sites are now unavailable because brickyards have been so generally abandoned and because of general topographical changes. At many of the sites visited, extensive excavation was necessary to reach fresh and undisturbed clay, especially at natural slides on rivers or brook ravines.

At each site visited in the present work, a vertical surface of clay was

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exposed from which the thickness of each varve was obtained for comparison with Antevs' data. Having established the dating of a site, samples were taken at approximately five-year intervals, using separate cups for each sample.

The cups are of bakelite-impregnated paper, in the form of a cylinder 1⅝ inches (4.12 cm) outside diameter and 1⅝ inches (4.12 cm) long, capped with a disc of clear plastic on which is scribed a reference-line across a diameter. On this line, near the edge of the cup, a hole one-quarter inch in diameter is provided which acts as a positioning hole when the cup is inserted in the measuring apparatus. The bottom edge of the cup is bevelled to provide a cutting edge to facilitate penetration into the clay when sampling. This end of the cup may be closed after sampling by cementing on a clear plastic disc to prevent drying out of the sample.

The bakelite-impregnated paper was the only material found suitable for rigidity and non-magnetic properties, but absorption of moisture from the sample caused swelling of the cup and, if measurements were not made promptly after sampling, the cup became too large to be inserted in the measuring apparatus.

After identification of the varves at a site, the years according to Antevs' chronology were marked directly on the clay face. To begin the sampling, the horizontal surface of a varve was exposed; in most cases this was quite simple, since the varves would separate readily at a sandy boundary.

Many of the layers thus exposed were not truly horizontal, but had so-called "tilts" up to several degrees. The word "tilt" has been adopted here to describe the departure from horizontal which geologists call "dip" because in magnetic studies the word "dip" commonly refers to the inclination of the Earth's magnetic field at any point, and some confusion would therefore arise.

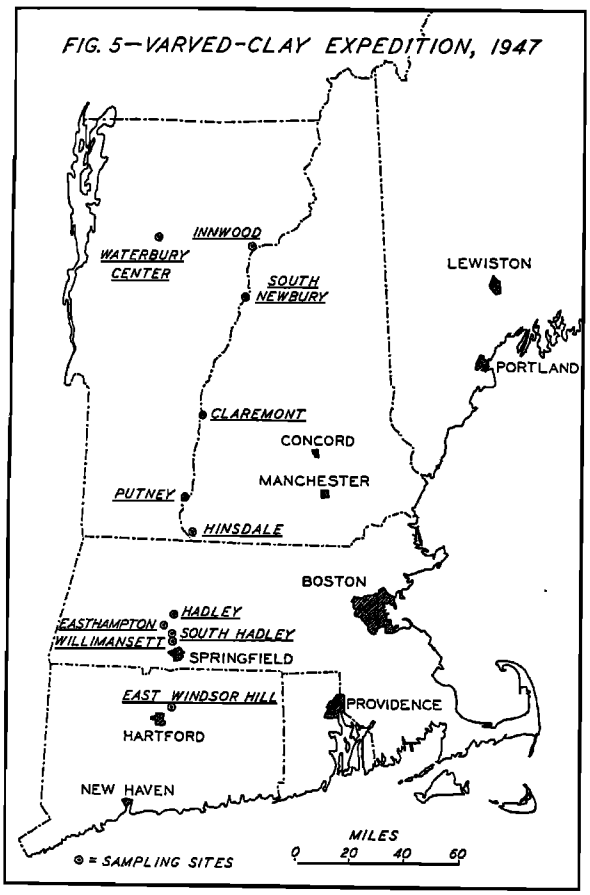
On the horizontal or nearly horizontal surface two lines were made, one in a magnetic north-south direction and the other at right-angles to the first. The tilts in these two directions were measured with a geologist's dip-indicator and recorded. A sample cup was placed on the surface with its scribed line parallel with the north-south line on the clay and with the positioning hole toward the north. A heavy brass disc, with a small, circular spirit level inset at its center, was placed on top of the cup and pressure applied to it to drive the cup into the clay until the cup was seen through the cover plate to be completely filled. The spirit level made it possible to keep the cup vertical while it entered the clay, within half a degree.

The problems arising from the taking of a vertical sample in a tilted bed are discussed later.

With the cup fully pressed in, the magnetic direction of the scribed line on the cup-cover was determined and recorded. The sample cup was then

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dug out of the clay, as shown in Figure 4, using brass tools, cleaned of adhering clay, and suitably identified. Measurements of the magnetic characteristics were made within a day or two. After measurement, the cups were emptied and used again, except for those containing samples spaced at 50-year intervals; these were sealed and retained for future reference.



If measurements on a particular sample were not consistent with those of adjacent samples, new samples were taken. Discrepancies were usually due to misreading of compass values and error in recording data. The technique of sampling was very satisfactory except when the clay was very hard or very soft.

Samples were sectioned to determine the extent to which the varves might be deformed during sampling. Very little distortion was found; at

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most it was confined to a two-mm annular ring in contact with the cup, and this was so small a part of the total volume that it could be neglected.

Results

Method of magnetization of clays—It is supposed that the sedimentary clays obtain their original polarization in the following way: The sediments in general contain only a very minute amount of magnetic material. In the case of the glacial clays deposited in the glacial lakes of New England, the magnetic material is principally magnetite. The magnetic content was determined by first separating out the small amount of magnetic material by stirring a suspension of a large amount of clay with an extremely strong magnet, then removing the magnetic powder from the magnet, and obtaining finally the X-ray diffraction pattern of the resulting material. This pattern corresponded within the accuracy of measurement to that of magnetite.

The size of the magnetic particles in these clays has not been determined exactly, but is too small to be resolved by an optical microscope. It is estimated that most of these particles have a diameter of less than 0.3 micron. There is therefore a strong possibility that these particles are single domains possessing an inherent magnetization, if Kittel's theory [15] is correct.

As these minute particles settle in still water, they are subjected to gravitational, thermal, hydrodynamic, and magnetic forces. The magnetic forces tend to align the particles in the direction of the Earth's magnetic field. Thermal forces tend to produce a random orientation of the particles and prevent any polarization of the clay. The gravitational and hydrodynamic forces tend to alter the direction of the particles, depending on their shape, away from an exact orientation in the direction of existing field of the Earth. It is intended to carry out a more complete investigation of these factors in the laboratory in the near future. There have not as yet been obtained sufficient satisfactory measurements on these factors to permit an adequate or careful discussion. Exploratory laboratory investigations do indicate that a satisfactory investigation of these factors can be carried out.

Laboratory deposition of clays—The laboratory investigation of glacial clays has been carried out by using a very thin suspension of clay in water, and depositing these clays in a controlled magnetic field within a non-magnetic container. The settling of the clay took place in a porcelain Buchner funnel. The first centimeter or so of clay was deposited by filtering under vacuum, but thereafter the filtering action was so slow that the clay particles settled out completely under gravity. Additional amounts of suspension were added carefully as the water was drawn away, thus avoiding the need for an impractical length of settling column. The filtering

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process reduced the time for deposition from several days to about 12 hours and assisted in the compacting and drying of the material. After rejecting the lower centimeter of deposit, cubical samples were cut from the remainder.

The results of measurements on these samples were as follows:

(*a*) The horizontal direction of polarization of the clays was always exactly the direction of the horizontal magnetic field within the experimental error of measurement.

(*b*) The angle between the horizontal and the magnetic vector of the polarization was always somewhat less than the angle of dip of the applied field. This difference was as much as 20° for weak fields but decreased progressively as the field strength increased.

(*c*) The magnetic polarization was a definite and reproducible function of the applied magnetic field.

(*d*) Measurements made at several different water temperatures indicated a change of polarization with temperature. Experimental difficulties in measurement have not as yet permitted an accurate determination of this temperature-coefficient, which in any event is small.

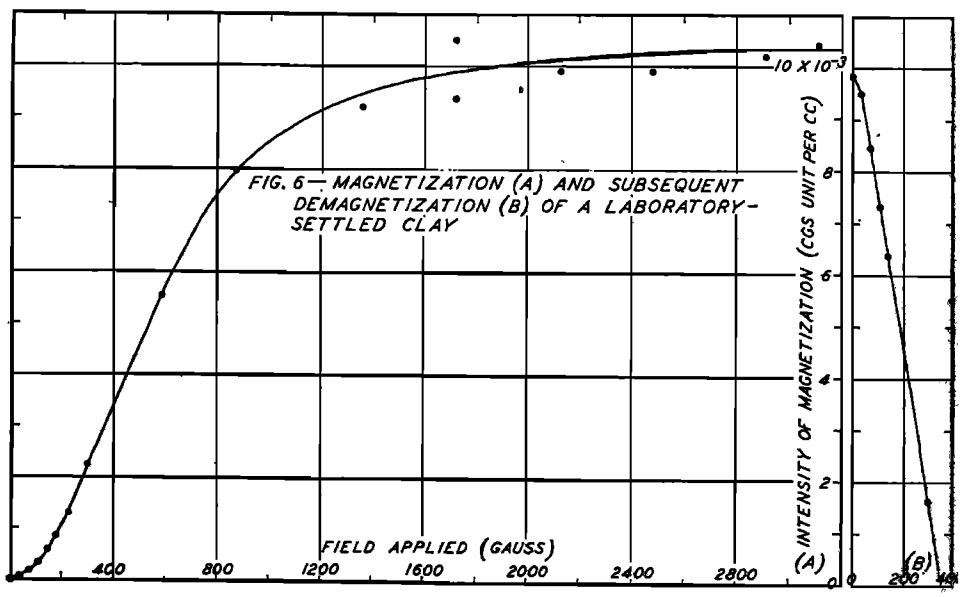


FIG. 6 — MAGNETIZATION (A) AND SUBSEQUENT DEMAGNETIZATION (B) OF A LABORATORY-SETTLED CLAY

From these measurements it is concluded that the initial polarization of the clays will faithfully reproduce the horizontal direction of magnetization but will only imperfectly reproduce the dip of the ambient magnetic field. This indicates that the shape of the particles affected by the gravitational and hydrodynamic forces during settling is important in determining

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the effect of the vertical component of the ambient field on the polarization. The existence of temperature-effect in the deposition experiments indicates that thermal forces may be important in an accurate determination of intensity of polarization as a function of the ambient magnetic field.

Magnetization curves of artificial sediments were taken. These are shown in Figures 6 and 7. Figure 6 shows the magnetization of a cube 1½ cm on the side deposited in zero magnetic field and later magnetized in uniform magnetic fields of increasing intensity and then demagnetized. Figure 7 shows the demagnetization curve of an artificial sediment de-

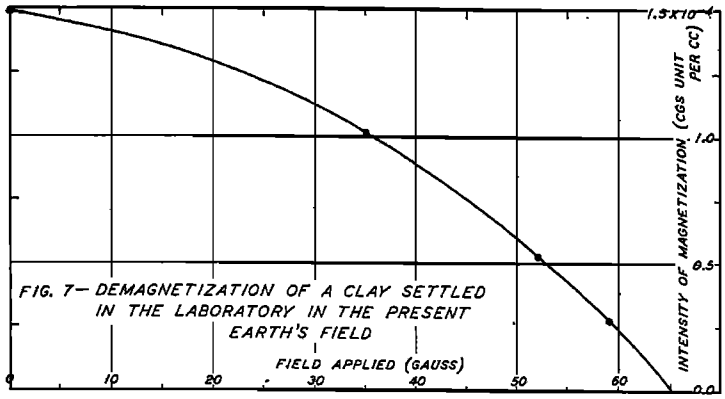


FIG. 7— DEMAGNETIZATION OF A CLAY SETTLED IN THE LABORATORY IN THE PRESENT EARTH'S FIELD

posited in the ambient field of the Earth. From these Figures it can be seen that a very large demagnetizing field of the order of 70 gauss is required to demagnetize a cube of the material. This shows the material has a high stability.

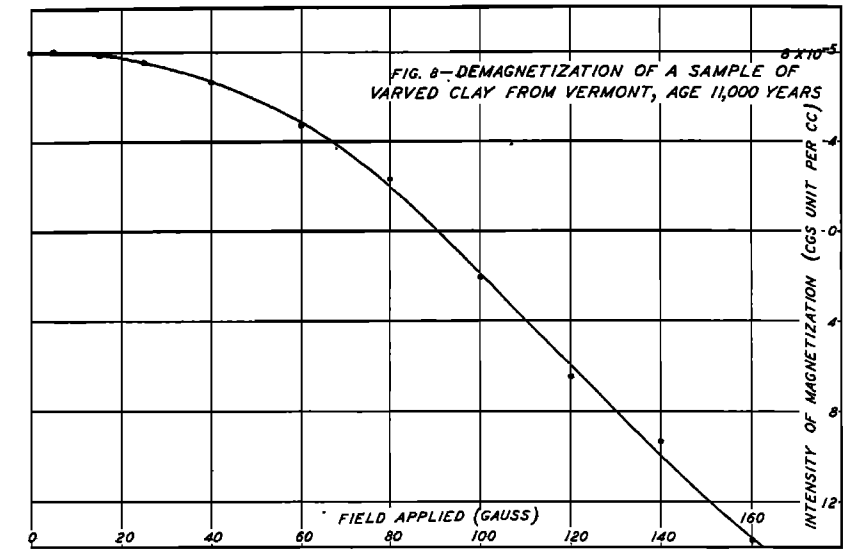
The demagnetization curve for a sample of the natural clay obtained from South Newbury, Vermont, is shown in Figure 8. A field of 90 gauss was required to demagnetize this sample, indicating again a high degree of stability of the material.

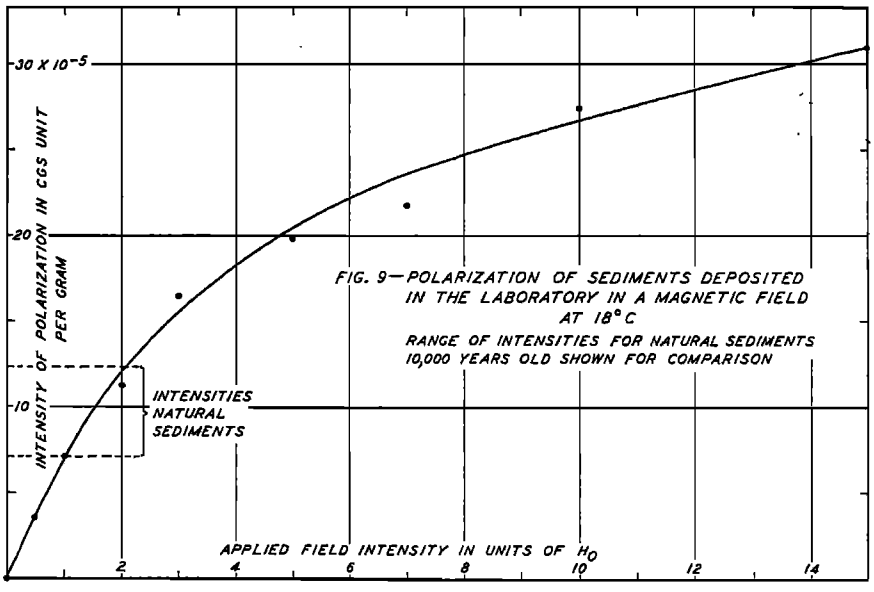
Samples of both natural and artificial varves were placed on a shaking table and shaken with amplitudes equivalent to very strong earthquakes over periods of 12 hours. This was the equivalent of many centuries of earthquakes in the New England region. The samples were placed in various possible orientations including that at right-angles to the ambient field of the Earth. The samples tested were both dry and very wet. No change in polarization within the experimental error was observed.

During one of the deposition experiments, the container holding the clays was accidentally rotated and the remaining clay deposited at a radically different angle. The lower clay in the deposit, although completely uncompacted at the time of shift of the container, retained the direction of the field prevailing at the time of deposition and did not change in the direction of the field after rotation of the container.

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All of these experiments indicate a very considerable stability of the polarization of the clays in so far as this could be determined by laboratory tests.





Intensity—One determination of intensity of the Earth's field at the time of deposition of the glacial clays has been made. The clay used was collected at South Newbury, Vermont, and represented deposition over a period of

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200 years. The polarization was measured and an average value for these years was obtained. Clay from the same years was thoroughly mixed with water and redeposited in the laboratory under varying field strengths. The resulting polarization of this clay as a function of field strength is given in Figure 9. The value of the ambient field corresponding to the natural polarization of this clay is of the same order as that of the present value of the Earth's field, within the experimental error of measurement. The age of this clay, in accordance with a preliminary determination due to W. D. Urry, is 10,000 years. Antevs estimated the age of the clay to be about 27,000 years.

According to this measurement, the intensity of the Earth's magnetic field has not changed more than one-half per cent per century, or about one-tenth the historical rate of decrease.

The intensity of magnetization of the New England clays varied, depending upon locality, about twentyfold. The maximum intensity observed was 40×10^{-5} cgs unit per cc and ranged down to 2×10^{-5} cgs unit per cc.

Probable error in measurement—Under field conditions, it was not always possible to obtain the maximum accuracy of measurement for which the apparatus was capable. The errors in measurement depended on the errors in sampling due to lack of verticality in taking samples, etc., as well as on errors in the apparatus. In any specific layer of clay one could also expect that there would be inhomogeneities, due to local currents acting during deposition, which would bring about some variation in the direction of polarization. This possibility was tested by sampling over large areas of the same layer.

Under the most favorable conditions when the clays were reasonably soft, the errors in sampling were very small indeed—less than the experimental errors of the apparatus. Under extreme conditions when the clay was nearly rock-like, the samples had to be taken by driving in the sampling cups under full body weight, and in such circumstance the sampling errors were often a number of degrees. Table 1 shows data for 20 samples taken from a single layer over a distance of about 100 feet (30 meters). These show a probable error in declination of about $\pm 2°$ for a single observation and a probable error in intensity of the horizontal component of polarization of ± 6 per cent.

Corrections to measurements—In the case of the glacial varves, corrections were made for the measured tilt of the layers. In specific cases it was suspected that these layers had been tilted after deposition. The measured tilts are given in Table 2. It was noted that there is a tendency for the layers to be tilted in the direction of the hillside when the clay is taken from the bank of a brook, ravine, or river bed. In fact, in extreme cases some deposits were rejected because of excessive slumping of the beds. However, in the

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TABLE 1—*Comparison of declination and intensity measurements from a single layer (Antevs 7006)*

Sample No.	Declination	Intensity
334	− 1°	71 × 10^{-5} cgs per cc
335	− 3.5	72
336	− 1.5	74
337	+ 2.5	78
340	− 5.5	74
341	+ 0.5	75
344	− 4.5	77
345	−11	79
348	− 1.5	72
349	− 3	72
352	− 4.5	66
353	− 3.5	63
356	− 0.5	..
357	+ 1	62
362	+ 1	83
363	+ 1.5	86
366	+ 1	75
367	+ 3	..
370	− 3	83
371	− 5	83
Mean	− 1.9	75 × 10^{-5} cgs per cc

Probable error of a single observation:
(*a*) In declination ±2°
(*b*) In intensity ±6 per cent

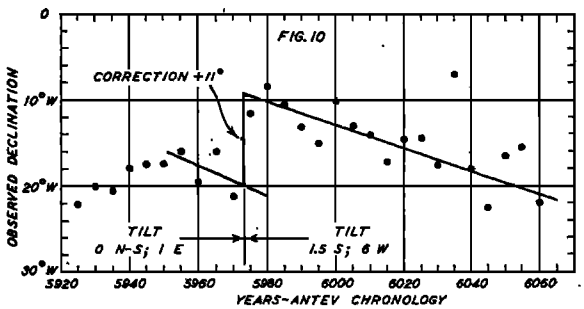
actual glacial lake there is no reason why the clay layers should be deposited in an exactly horizontal bed. Thus there is every possibility that the clays may have had an initial slope the same as that observed when they were collected. Faced with this dilemma, it was decided to correct all the measurements to that of an exactly horizontal layer. This was done by equation (1) which gives the results to a sufficiently good approximation.

$$\alpha = \tan I(T_n \sin D - T_e \cos D) \qquad (1)$$

Here α = angle of correction; I = angle of inclination or "dip" of Earth's magnetic field, positive with north pole downward; T_n = tilt of bed north (or south), positive when downward to the north; T_e = tilt of bed east (or west), positive when downward to the east; D = measured angle of declination, positive in a clockwise direction. In the process of sampling, as was explained earlier, the sample itself was always taken so that the axis of the cup was vertical and the index mark on the top of the cup approximately in the direction of the present declination. The east-west and north-south

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tilts of the layer were measured. Equation (1) corrects the declination values actually observed for the errors introduced by the tilts. The amount of this correction was usually small and in no case exceeded 12°. In general, the effect of this correction was to make the observed polarization for the same years more consistent, although some anomalies were encountered. Figure 10 shows a typical improvement in matching results from two distinct locations resulting from the application of corrections for tilt of the



EFFECT OF TILT (NON-HORIZONTALITY) OF VARVES ON OBSERVED VALUES OF DECLINATION, PUTNEY, VERMONT, ANTEV STATION NO. 31

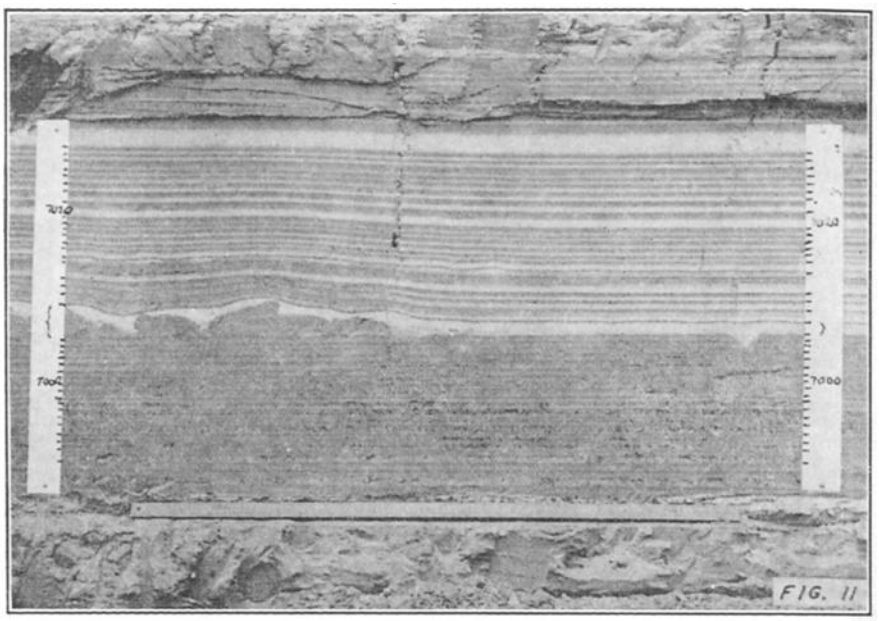
Place	Station No.	Years	Tilt of Varves		Sample cup numbers	Samples taken
			N-S	E-W		
	(Antevs)	(Antevs)				
SOUTH NEWBURY (BRADFORD), VT.	73	6950-8450	0° to 1° S	0° to 1/2° W	1-401	401
INWOOD, VT.	88	7256-7292 7312-13	6° S to 6° W		403-412	10
WATERBURY CENTER, VT.	151	7122-7176	.....	.....	413-423	11
CLAREMONT, N. H.	63	6665-6950 6950-7000	1/2° to 8° S ±20° E-W	1° W to 1° E ±20° E-W	424-486 487-504	63 18
PUTNEY, VT.	31	5715-6235	0° to 3° S	1° E to 6° W	505-635	131
HINSDALE, VT.	29	5470-5480 5576-5590	....	....	636-643	8
HADLEY, MASS.	23	5025-5470	2° N	3° W	644-737	94
SOUTH HADLEY FALLS, MASS.	11-14	4050-4120 4210-4275	1° S 1° N	2-1/2° E 1-1/2° E	738-771	34
EASTHAMPTON, MASS.	18	4615-5060	1-1/2° N to 6° S	1° E to 5° N	772-855	84
WILLIMANSETT, MASS.	10	4105-4165	.....	.....	856-869	14
EAST WINDSOR HILL (HARTFORD), CONN.	6	3040-3730	0°	1° W to 2-1/2° E	871-1021	151
TOTAL SAMPLES						1019

TABLE 2—SUMMARY OF VARVE-SAMPLING PROGRAM IN NEW ENGLAND, SEPTEMBER TO NOVEMBER, 1947

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beds. It is notable that clays from widely separated localities gave consistent results for declination.

Stability—Stability of the polarization of the glacial varves at South Newbury was investigated in the following way. Between the years 7006 and 7008, Antevs' chronology, an anomalous deposit of clay was found in excavating, as shown in Figure 11. This material consisted of varved clay



which had been mixed and kneaded, and then deposited with the layers randomly oriented. This abnormal material existed in patches. Since the layers below these patches and the layers above coincided exactly with the normal series existing in nearby exposures, it is firmly established that this clay was deposited in some special way, perhaps by dropping off of an iceberg floating in the glacial lake. Thus the disturbance in the layer occurred at the time of the deposition, and any magnetic irregularities in this layer as compared with the normal layers have persisted over the entire period since the time of deposition.

If the polarization in the normal layers is unstable and has changed so as to follow the ambient field of the Earth in direction and intensity, then the polarization of the disturbed layer should, during the last 10,000 years, have also been magnetized substantially in the same direction and with the same intensity as the normal layers. On the other hand, if it had retained a more or less statistical magnetization, as would be indicated by its stirred and mixed nature, then the magnetic direction and intensity of

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this clay should be much more nearly statistical. The intensity, for example, would be expected to go from nearly zero to approximately the same value as the normal layers. Figure 12 shows the plot of 20 samples taken from

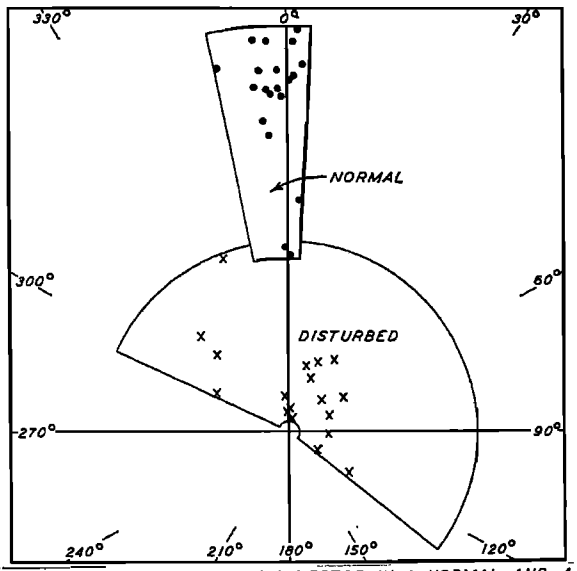


FIG. 12—SCATTER OF MAGNETIC VECTOR IN A NORMAL AND A DISTURBED LAYER, NEW ENGLAND VARVED CLAY, 1947

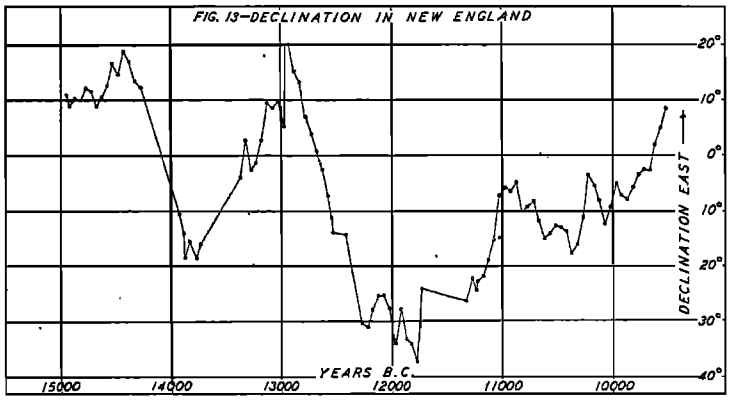
the disturbed layer of clay and 20 samples taken from the preceding year. This Figure shows conclusively that the disturbed layer has retained over geologic time a substantially random polarization as compared with the normal layer. It is thus concluded that the glacial clays at South Newbury have substantially retained the polarization which they obtained when they originally settled in the glacial lake. In an unpublished doctorate research carried out by John W. Graham for submission to the Department of Geology of Johns Hopkins University, additional evidence is presented to show that the magnetic vector in a tilted layer in the New England clays remains very nearly unaffected by tilt.

Declination—The results for declination are given for the glacial varves in Figure 13. These cover a period of about 5000 years. The absolute dating given is based on tentative results by W. D. Urry, the relative dating from Antevs. Each point in Figure 13 corresponds to an average taken over 50 years. The number of samples or years actually measured for each point varies from 5 to 15. One thousand measurements are included in the entire curve.

Over the period of 5000 years, it can be seen that the declination remained substantially constant on the average, with excursions of the order

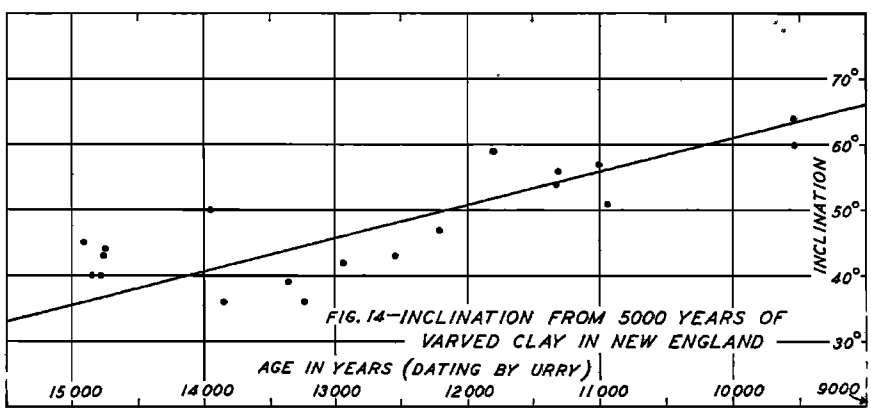
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of ±30°. These excursions are consistent with the measurements made at London, Boston, and Baltimore during the last few hundred years. The average value of declination over the 5000 years is +10° from the present declination, which is about −15° in New England. There is some slight



suggestion of a periodicity of the order of about 300 years. The authors do not regard this as having any particular significance since the number of cycles is too few to permit an evaluation of the reality of such a cycle.

Inclination—The authors do not have as yet much confidence that the values of inclination measured in the glacial varves have an exact relation to the inclination at the time of deposition. These doubts arise largely from the results of the laboratory settling of the clays. For the sake of completeness, however, the inclination measured is given in Figure 14.



Preliminary measurements from Pacific Ocean cores

Some preliminary results for a much larger time-scale have been obtained from a study of magnetic polarization of Pacific Ocean bottom

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sediments, based on samples obtained on the United States Antarctic Expedition of 1946-47. Through the courtesy of the United States Navy Hydrographic Office and J. L. Hough, who as a member of the expedition obtained the ocean-bottom samples, six cores were supplied to the authors in 1947. Five of these cores have been partially studied. The geographical location of each core and its length are given in Table 3.

TABLE 3—*Pacific Ocean bottom samples obtained on United States Antarctic Expedition of 1946-47*

Sample No.	Latitude	Longitude	Depth	Remarks	Length of core
	° ′	° ′	*fathoms*		*feet*
1	08 56 S	92 05 W	2150	Distinct stratification; alternately red clay & glob. ooze	6½
2	32 21 S	105 55 W	1980	Mainly red clay	5
3	68 26 S	179 35 W	1800	Glacial marine	8
4	69 12 S	179 34 E	2040	Glacial marine	8½
5	70 17 S	178 23 W	1635	Glacial marine	8½
10	44 14 S	175 15 E	300	Greenish-gray mud	5

It is assumed that at the bottom of the Pacific Ocean conditions similar to those in the glacial lakes prevail and small amounts of magnetic minerals are being deposited along with the well-known deep-sea sediments. So far it has not been determined what these magnetic minerals are. They are most probably iron compounds, since the intensity of magnetization increases with the proportion of red clay in the samples, while some cores which are composed mostly of globirgerina ooze are almost non-magnetic. It is possible that the magnetic material of the cores is the magnetic variety of ferric oxide which differs from the ordinary variety in having a cubic lattice. Whatever the material is, it is present in such minute quantities that it is hard to separate and identify.

In the case of the glacial clays, it has been shown that their polarization has been retained since they were formed on the lake bottom, and that they cannot become polarized by induction in the subsequent Earth's field. Parallel experiments on the deep-sea cores have not been carried out, since these require a set of fortuitous circumstances not granted so far. Thus it cannot be positively stated that the polarization they obtained at the time of deposition is the same as that now measured.

The previous work of McNish and Johnson [7*a*] was done on a sea core obtained in the Atlantic by Piggot. The results obtained agreed with those of the varve clays of New England.

The Pacific cores were received sealed in the plastic tubes in which they

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were taken. They measured approximately 1½ inches in diameter. Each tube was slit horizontally into halves for its entire length. One-half of each core was returned to Professor Hough for his own studies, the other half was used by the authors for magnetic studies and by Dr. W. D. Urry for dating the sediments by measurements of radioactive equilibrium. The results obtained by Dr. Urry are used in this paper. His methods and procedures in dating the samples are discussed elsewhere.

Dr. Urry's samples were taken from each core first. The remaining material was cut up into cubes, taken successively throughout the length of the core, each cube being 1½ cm on the side. Each cube was suitably identified and covered with a clear lacquer to prevent loss of moisture and

TABLE 4—*Values of magnetic inclination*[a] *from Pacific Ocean cores*
(Sample identification in centimeters from top of core)

Core									
N-1		*N*-2		*N*-3		*N*-4		*N*-5	
Sample	Observed *I*	Sample	Observed *I*	Sample	Observed *I*	Sample	Observed *I*	Sample	Observed *I*
	°		°		°		°		°
88- 90	−43	24- 26	−21	10- 12	−74	3- 5	−47	26- 28	−78
94- 96	−22	51- 53	−24	26- 28	−63	24- 26	−61	100-102	−83
132-134	− 2	88- 90	−19	44- 46	−62	54- 56	−54	132-134	−93
134-136	2	90- 92	−17	62- 64	−73	62- 64	−47	178-180	−70
140-142	2	102-104	− 8	78- 80	−64	100-102	−63		
156-158	−25	104-106	− 7	84- 86	−60	122-124	−56		
				102-104	−66	132-134	−59		
				112-114	−73	144-146	−56		
				142-144	−49	158-160	−65		
				188-190	−39	196-198	−64		
				214-216	−21	220-222	−59		
						230-232	−59		
						254-256	−46		
Sum	−88	.....	−96	.....	−644	.....	−736	.....	−324
Mean	−15	.....	−16	.....	−59	.....	−57	.....	−81
Present value	0	.....	−42.5	.....	−83	.....	−83	.....	−83

[a]Inclination is negative when south pole dips downward.

disintegration. From the five cores, approximately 400 cubes were obtained.

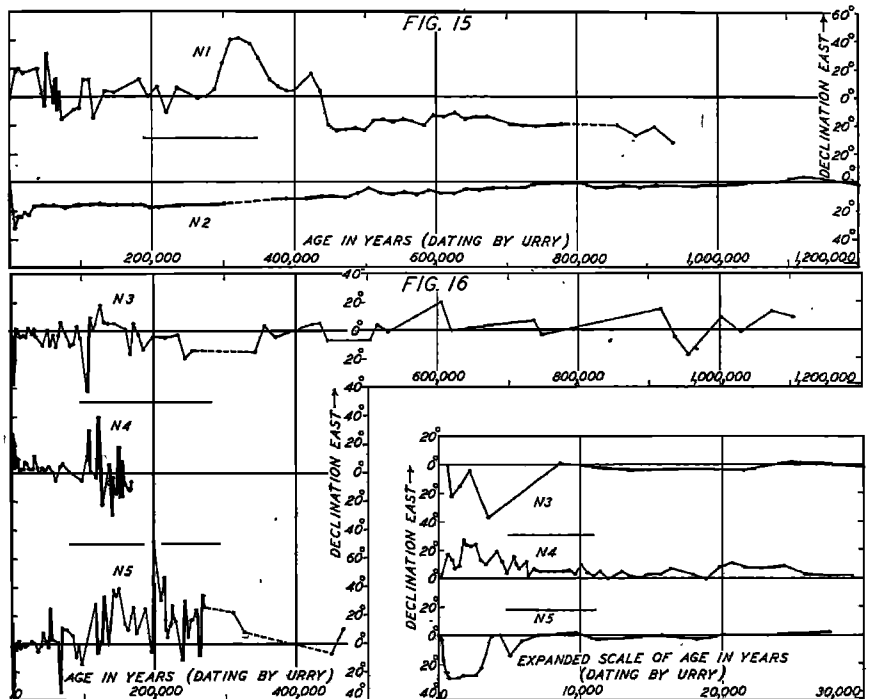
The measurement of most interest is the variation in direction of hori-

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zontal component of polarization, which, it is assumed, represents the variation in declination of the Earth's field when the core material was formed on the ocean floor. Cores 1 and 2 were taken close to the magnetic equator, as shown in Table 3, and hence the horizontal component is quite strong. The direction of the horizontal component in these could be measured with higher accuracy than in the case of the remaining three cores. The latter were taken close to one another in the south polar regions and are polarized almost vertically; that is, along the axes of the cores (see Table 4).

In Table 4, the inclination of polarization of randomly selected cubes from all five cores is given, along with the actual magnetic inclination of the Earth's field at the present day at each location. The agreement between the two sets of values is fair.

The results for the declination obtained from cores 1 to 5 are given in Figures 15 and 16. Core 1 shows the biggest range in values which is com-



SECULAR VARIATION IN MAGNETIC DECLINATION FROM CORES OF PACIFIC OCEAN SEDIMENTS FROM UNITED STATES ANTARCTIC EXPEDITION 1946-47

parable to measurements of deviation made in historic times at London and Boston. Core 2 has a remarkably small scatter of values, but there is a gradual but steady change in declination throughout. The trend shown may be real, but too much reliance cannot be placed in it, since it represents

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only a small difference of angle between samples taken at the top and the bottom of the core, and this could have arisen from rotational distortion during sampling.

In this work, it is assumed that the top of the core is magnetized in the same direction as the present Earth's field, which gives the zero value in Figures 15 and 16. The intensity of magnetization varied from 3×10^{-6} to 3×10^{-5} cgs unit per cc.

In order to determine whether the sediments could be magnetized in fields of the order of the present Earth's field, some sample cubes were subjected to magnetic fields in the range of 1 to 100 gauss, the direction of the field being opposed to the natural polarization. The field in each case was increased from zero up to the value plotted, held there for a few seconds, and then decreased to zero. The intensity of the polarization of the sample was then measured. Successive values of the applied field were increased until the polarization was at least as great as formerly but opposite in direction. The results show that fields quite large compared to the Earth's field have to applied before the polarization is reduced to zero. Values of this applied field ranged from 2 to 40 gauss, and for each sample the demagnetizing field was directly proportional to the natural polarization of the sample.

An attempt was made to find out from the sea deposits how the Earth's field varied in intensity over long periods. Core *N*2 was chosen for experiment for several reasons. It was almost uniformly composed of red clay; its polarization was quite strong, on the average (about 2×10^{-5} cgs/cc); its plane of polarization was almost horizontal.

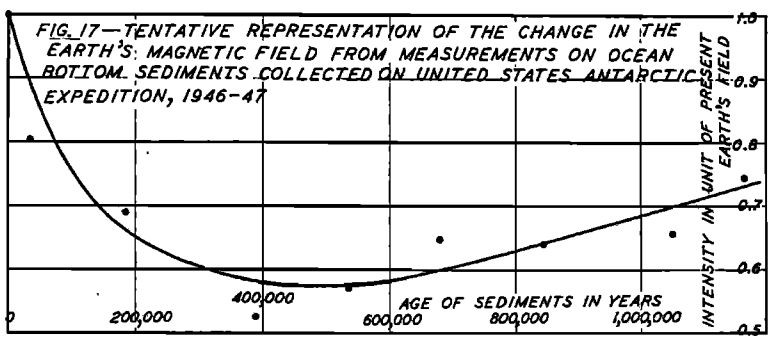
The intensity of polarization of several samples was measured. These samples were then brought into and after a few seconds removed from a field of about 2000 gauss. The intensity of polarization was again measured. On the assumption that the magnetic particles are the same size throughout, which is most probably true, then the latter measurements are proportional to the quantity of magnetic material present. This gives a basis for comparison of the original natural polarizations, which in turn should be proportional to the strength of field at the time of deposition. The results are shown in Figure 17. The significant points are the small change throughout the one million years of sedimentation and the good agreement between the readings. There is no large change in the direction of the polarization throughout the cores and it is concluded that one million years ago the Earth had a magnetic field similar in direction and size to the present field and that the south pole was in approximately the same position as at the present time.

In connection with the measurements of declination, it was noted that in spite of the necessarily random orientation of the reference face in each core, the measurements had a tendency to fall in a very limited range of

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orientations from core to core. That all five cores should exhibit this tendency would seem to be a remarkable coincidence.

That the cores were exposed to large demagnetizing fields does not appear probable; neither do there appear to have been any systematic factors in storage or transportation capable of explaining the observed results. Finally, the measurements of inclination would have been considerably altered and possibly have been made similar in all cores by any artificial demagnetization mechanisms. The observed inclinations are different in each core and are consistent with the geographical location of



the sample. It is concluded that the observed coincidence must be fortuitous, but the authors reserve final judgment on the results for these cores because of this circumstance.

Summary of conclusions

In the earlier work reported by Johnson and McNish and by Ising, there appeared to be a very great uncertainty with respect to the stability and interpretation of the magnetization of glacial clays and ocean sediments. The data presented in this paper indicate that in the case of the glacial clays the polarization of these clays has remained constant in direction and intensity since the clays were deposited 10,000 to 20,000 years ago. This conclusion is reinforced by laboratory experiments in redepositing the clays and testing their stability as well as by their high coercive force. The consistency of the results for the ocean-bottom sediments makes it probable or at least possible that these sediments may also have remained constant in polarization since they were deposited.

It can be tentatively concluded that the direction of the Earth's magnetic field has remained substantially constant in horizontal direction during the last million years, and the intensity has also remained substantially constant, within a rather large experimental error, during the last million years. Further work is under way to extend the period of measurement to times of the order of one hundred million years and to determine

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more accurately, by laboratory investigations, the variation in intensity of the Earth's magnetic field with time.

The rate of change in intensity of the Earth's magnetic field is shown to have been less than one-half per cent per century during the last 10,000 years and probably to have been less than one-hundredth per cent per century during the last million years.

The conclusions stated above have a particular importance with respect to the "fundamental" theories proposed by Schuster and Blackett. Since there is no evidence that the rotation of the Earth has varied in direction or magnitude, a very great variation in direction or intensity of the Earth's magnetic field with time would invalidate the theories proposed by Schuster and Blackett. However, no great variation has been found from the present work, and the results are consistent therefore with the "fundamental" theory. On the other hand, no positive evidence is provided by this investigation to indicate that the fundamental theory is correct.

The authors appreciate the very considerable assistance given to them in making the necessary laboratory measurements by P. F. Michelsen of the Department, Miss Juliet Rocquet of the Institut de Physique du Globe, and Dudley Parkinson, now a graduate student at the Johns Hopkins University.

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